Total No. of Questions : 8]

SEAT No. :

PE-2169

[Total No. of Pages : 3

[6584]-68

B.E. (Civil)

Transportation Engineering

(2019 Pattern) (Semester - VII) (401002)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates :

1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
2) *Figures to the right indicate full marks.*
3) *Use of logarithmic tables, slide rule, Mollies charts, electronics pocket calculator and steam tables is allowed.*
4) *Assume suitable data, if necessary.*
5) *Neat diagrams must be drawn wherever necessary.*

Q1) a) A car travelling at 22.22 m/sec is overtaking another car moving at 60 kmph on a two lane undivided highway. Assuming an acceleration of the overtaking car as 0.7 m/ sec^2; Calculate **[10]**

i) minimum overtaking sight distance; and

ii) minimum and desirable length of overtaking zones

b) State the various objects of providing extra widening on horizontal curves. **[6]**

c) Distinguish between carriageway width and formation width. **[2]**

OR

Q2) a) Calculate the length of transition curve required for a two lane carriageway of 7.0 m width on horizontal curve, if the design speed is 65 kmph and the road is passing through rolling terrain. The radius of horizontal curve is 200 m. Take extra widening of 0.66 m **[10]**

b) Define gradient. State the various types of gradients. Also state the values of gradient recommended by IRC for plain and rolling terrain. **[6]**

c) State the methods of elimination of crown while providing the superelevation in the field? **[2]**

P.T.O.

Q3) a) Mention the step by step procedure of 'Ductility test on bitumen' **[6]**

b) Define Flakiness Index (FI). How FI is determined in the laboratory. **[6]**

c) List the different types of cutback bitumen. When are these used **[5]**

OR

Q4) a) For a construction of a bituminous road in a certain locality, contractor has received 60/70 grade of bitumen from refinery? Explain in brief the test to be carried out to confirm the adhesion property of bitumen. **[6]**

b) Define Elongation Index (EI). How EI is determined in the laboratory. **[6]**

c) What are the different methods of classifying / grading the bitumen? Explain in brief. **[5]**

Q5) a) Using the following data calculate the wheel load stress at the interior region of a cement concrete pavement using H.M. Westergaard's equation: **[8]**

Wheel load = 5100 kg, Modulus of Elasticity of concrete = 3×10^5 kg/cm^2, Pavement thickness = 15 cm, Poisson's ratio = 0.15, Modulus of subgrade reaction =7.0 kg / cm^3, Radius of wheel load contact = 16 cm.

b) Draw a neat labelled sketch of cross section of flexible pavement and rigid pavement. **[6]**

c) Write a note on temperature stresses in concrete pavement. **[4]**

OR

Q6) a) A two lane two way road is carrying an initial traffic of 1200 Commercial Vehicles per day (CVPD) is to be strengthened to cater the need of growing traffic. The VDF has been found to be 2.0. The rate of growth of traffic is 7.5% per annum. The pavement is to be designed for 15 years. Calculate the cumulative standard axles to be used in design. It is suggested to use the factor of 1.0 to account for lateral placement of wheel loads. **[8]**

b) Mention the step wise procedure of design of flexible pavement as per IRC guidelines. **[6]**

c) Explain in brief PQC and DLC. **[4]**

Q7) a) A bridge has linear waterway of 160 meters constructed across a stream whose natural linear waterway is 230 meters. If the average flood discharge is 1250 m^3 per seconds and average flood depth is 3.5 meters, calculate afflux under the bridge. **[6]**

b) Derive an expression for economical span of the bridge. **[6]**

c) State the various requirements of an ideal permanent way. **[5]**

OR

Q8) a) Name the different types of loading while designing a brief. Explain in brief. **[6]**

b) State the various ideal bridge site characteristics. **[6]**

c) Define gauge. State and explain various gauges adopted by Indian railways. **[5]**

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